

Percent Change from Old Price to New Price

Spelled out methods...

Remember, use decimal form of percentages in calculations. For example, an increase of 23% means `percent_increase` is 0.23 and the `scale_factor` is 1.23.

Find `new_price` from `old_price` and `percent_increase`

```
scale_factor = 1 + percent_increase  
new_price = old_price * scale_factor
```

Find `new_price` from `old_price` and `percent_decrease`

```
scale_factor = 1 - percent_decrease  
new_price = old_price * scale_factor
```

Find `old_price` from `new_price` and `percent_increase`

```
scale_factor = 1 + percent_increase  
old_price = new_price / scale_factor
```

Find `old_price` from `new_price` and `percent_decrease`

```
scale_factor = 1 - percent_decrease  
old_price = new_price / scale_factor
```

Find `percent_increase` from `old_price` and `new_price`

```
scale_factor = new_price / old_price  
percent_increase = scale_factor - 1
```

Find `percent_increase` from `old_price` and `new_price`

```
scale_factor = new_price / old_price  
percent_decrease = 1 - scale_factor
```

Formulas with variables

s = scale factor

i = initial (old) price

n = new price

c = percent change (decimal form)

$$s = \frac{n}{i}$$

$$i = \frac{n}{s}$$

$$n = si$$

$$c = |s - 1|$$

- If new price is **more** than initial price, the percent change is a percent **increase**, and $s = c + 1$, and $c = s - 1$, and:

$$c = \frac{n}{i} - 1$$

$$i = \frac{n}{1 + c}$$

$$n = i \cdot (1 + c)$$

- If new price is **less** than initial price, the percent change is a percent **decrease**, and $s = 1 - c$, and $c = 1 - s$, and:

$$c = 1 - \frac{n}{i}$$

$$i = \frac{n}{1 - c}$$

$$n = i \cdot (1 - c)$$